

Synthesis and Characterization of Potassium Tris(oxalato)ferrate(III) Trihydrate

A Spectrophotometric Method of Iron Analysis

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We have adopted the potassium tris(oxalato)ferrate(III) trihydrate ($\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$, potassium ferrioxalate) empirical formula experiment (1) for our general chemistry laboratory and find it, as advertised, to be an excellent integrative experience in synthesis and characterization for the students. The synthesis and recrystallization of the title compound proceed smoothly, and we follow the prescribed methods for the analytical determination of oxalate (by KMnO_4 titration) and waters of hydration (by gravimetry). However, we have introduced a fast and accurate spectrophotometric method for the determination of iron in the product that takes the place of the photochemical–gravimetric procedure described in Reference 1. Besides the pedagogic interest of bringing three different types of chemical analysis (titrimetric, gravimetric, spectrophotometric) to bear on one compound, the new iron determination allows us to complete the experiment (with students working in pairs) in two three-hour laboratory periods rather than the five periods allotted in Reference 1.

The iron content of the potassium ferrioxalate product is determined by making the tris(bipyridine)iron(II) complex ion directly from the $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ complex ion and measuring the absorbance of the resulting $[\text{Fe}(\text{bpy})_3]^{2+}$ solution. Approximately 0.15 g (weighed to the nearest 0.001 g) of recrystallized and air-dried potassium ferrioxalate trihydrate product is dissolved in deionized water, quantitatively transferred to a 100-mL volumetric flask, and diluted to the mark with DI water. Using a transfer pipet, 1.00 mL of this solution is placed in a 50-mL volumetric flask and about 10 mL of DI water is added. The following reagents are then added to the 50-mL volumetric flask in order (2): 10 mL of 0.1% aqueous bipyridine solution, 2 mL of 10% aqueous hydroxylamine solution (to reduce the Fe(III) to Fe(II)), and 5 mL of 2 M sodium acetate (to buffer the solution in the 5–7 pH range). The solution is diluted to the 50-mL mark with DI water and mixed thoroughly. The absorbance of the solution is measured at 522 nm. We use a Hewlett–Packard 8452A diode array spectrophotometer and a 1.00-cm path length cuvette, although any simple single-beam spectrophotometer could be used provided that either the cell path length is known or a calibration curve of A_{522} versus $[\text{Fe}(\text{bpy})_3^{2+}]$ is generated. Using Beer's Law and the solution volumes, the amount of Fe (in mmol) in the ~0.15 g sample of $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$ product can be determined as follows:

$$\frac{\text{mmol Fe}}{\text{g product}} = \frac{A_{522}}{a \times b} \times 50.0 \text{ mL} \times \frac{100 \text{ mL}}{1.00 \text{ mL}} \times \frac{1}{m_{\text{sample}}}$$

where a = molar absorptivity of $[\text{Fe}(\text{bpy})_3]^{2+}$ at 522 nm = $8650 \text{ M}^{-1}\text{cm}^{-1}$ (2), b = cell path length (in cm) and m_{sample} = mass of potassium tris(oxalato)ferrate trihydrate sample 0.15 g. The percentage of Fe in $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$ can be calculated by converting the above result to (g Fe/g product) and multiplying by the atomic mass of iron.

The theoretical value for Fe in the potassium ferrioxalate trihydrate product can be expressed as either 2.036 mmol Fe per gram of $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$ or as 11.37% Fe in $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$ by weight. In our general chemistry laboratories over the past two years, 44 student pairs obtained an average iron result of 2.030 mmol Fe/g $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$ (11.34% Fe), with an relative standard deviation of 12.0% and an average error from the theoretical value of less than 1%. The results of 27 of the 44 student pairs were within 5% of the theoretical value and 37 of the 44 pairs were within 10% of the theoretical value. Thus, the spectrophotometric iron determination is comparable in quality to the much lengthier method outlined in Reference 1 (photolytic reduction to and gravimetric determination of ferrous oxalate) and is accurate enough to get good empirical formula results. The major sources of error in the spectrophotometric analysis of iron are probably insufficient drying of the potassium ferrioxalate trihydrate product (the spectrophotometric iron analysis was carried out during the same lab period as the synthesis) and imprecise use of the small 1-mL transfer pipet in the hands of first-semester chemistry students.

We have retained the photolytic reduction of the ferrioxalate solution as a qualitative exercise for the students to perform while their ferrioxalate product is drying for the analytical procedures. Irradiation of the green ferrioxalate solution (approximately 0.5 g in 5 mL of 10% by volume acetic acid) with either a high intensity tensor lamp or a slide projector forms a yellow solution, a white ferrous oxalate precipitate and bubbles of CO_2 , in less than 10 min. Students are asked to record and describe their observations of this photochemical process as a function of irradiation time.

Literature Cited

1. Olmsted, J. J. *J. Chem. Educ.* **1984**, *61*, 1098–1099.
2. Moss, M. L.; Mellon, M. G. *Ind. Eng. Chem., Anal. Ed.* **1942**, *14*, 862–865.